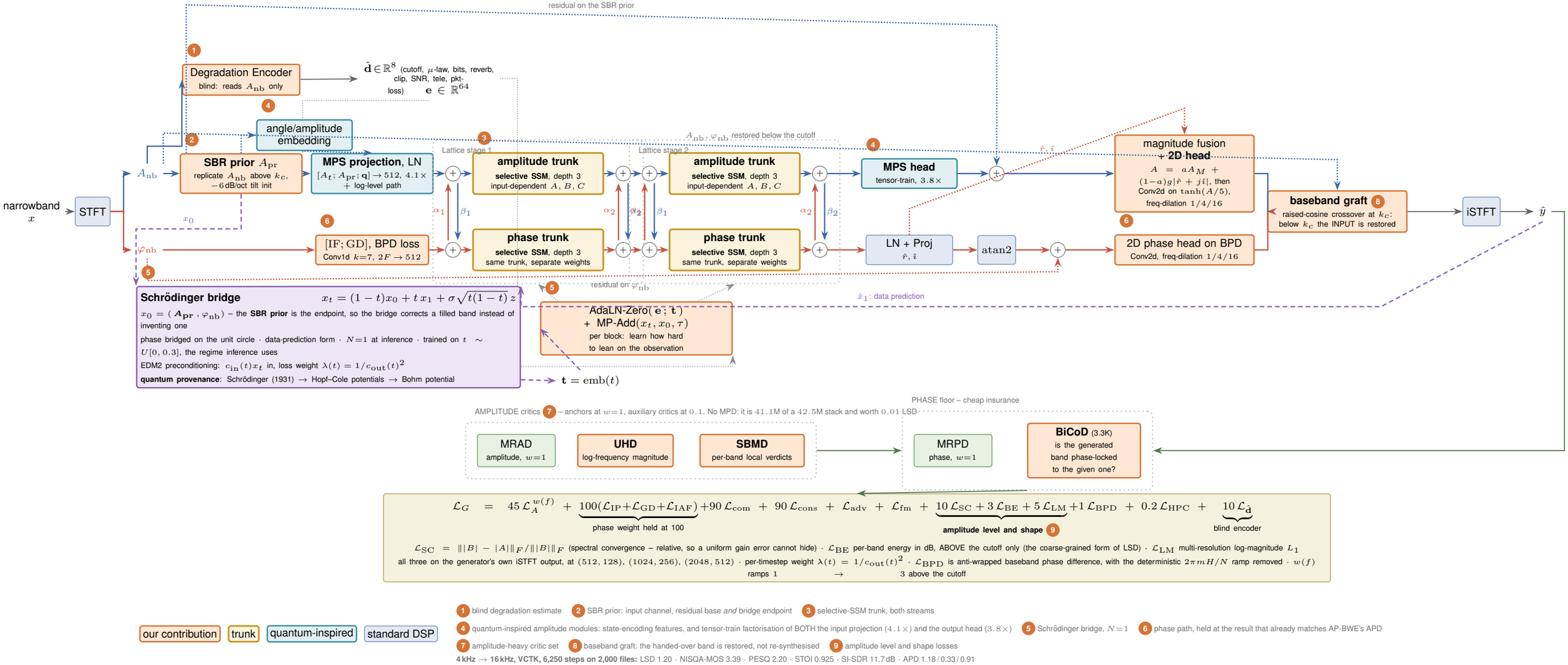

residual on the SBR prior
1
Degradation Encoder
blind: reads A_{nb} only
$\hat{\mathbf{d}} \in \mathbb{R}^8$ (cutoff, μ-law, bits, reverb, clip, SNR, tele, pkt-loss) $\mathbf{e} \in \mathbb{R}^{64}$
4
angle/amplitude embedding
2
SBR prior A_{pr}
replicate A_{nb} above k_c, -6 dB/oct tilt init
MPS projection, LN
$[A_t; A_{\mathrm{pr}}; \mathbf{q}] \to 512$, $4.1\times$ + log-level path
3
Lattice stage 1
amplitude trunk
selective SSM, depth 3
input-dependent A, B, C
Lattice stage 2
amplitude trunk
selective SSM, depth 3
input-dependent A, B, C
$A_{\mathrm{nb}}, \varphi_{\mathrm{nb}}$ restored below the cutoff
4
MPS head
tensor-train, $3.8\times$
narrowband x
STFT
A_{nb}
x_0
φ_{nb}
6
[IF; GD], BPD loss
Conv1d $k{=}7$, $2F \to 512$
phase trunk
selective SSM, depth 3
same trunk, separate weights
phase trunk
selective SSM, depth 3
same trunk, separate weights
α_1 β_1 α_2 β_2 α_2 β_1 α_2 β_2
LN + Proj
$\hat{r}, \hat{i}$
atan2
$\hat{r}, \hat{i}$
magnitude fusion + 2D head
$A = aA_M + (1-a)g|\hat{r} + j\hat{i}|$, then Conv2d on $\tanh(A/5)$, freq-dilation $1/4/16$
6
2D phase head on BPD
Conv2d, freq-dilation $1/4/16$
baseband graft 8
raised-cosine crossover at k_c: below k_c the INPUT is restored
iSTFT
$\hat{y}$
5
residual on φ_{nb}
$\hat{x}_1$: data prediction
Schrödinger bridge
$x_t = (1-t)x_0 + t\,x_1 + \sigma\sqrt{t(1-t)}\,z$
$x_0 = (\boldsymbol{A}_{\mathbf{pr}}, \varphi_{\mathrm{nb}})$ – the SBR prior is the endpoint, so the bridge corrects a filled band instead of inventing one
phase bridged on the unit circle · data-prediction form · $N{=}1$ at inference · trained on $t \sim U[0, 0.3]$, the regime inference uses
EDM2 preconditioning: $c_{\mathrm{in}}(t)x_t$ in, loss weight $\lambda(t) = 1/c_{\mathrm{out}}(t)^2$
quantum provenance: Schrödinger (1931) → Hopf–Cole potentials → Bohm potential
AdaLN-Zero($\mathbf{e}$; $\mathbf{t}$) + MP-Add(x_t, x_0, τ)
per block: learn how hard to lean on the observation
$\mathbf{t} = \mathrm{emb}(t)$
PHASE floor – cheap insurance
AMPLITUDE critics 7 – anchors at $w{=}1$, auxiliary critics at 0.1. No MPD: it is 41.1M of a 42.5M stack and worth 0.01 LSD
MRAD
amplitude, $w{=}1$
UHD
log-frequency magnitude
SBMD
per-band local verdicts
MRPD
phase, $w{=}1$
BiCoD (3.3K)
is the generated band phase-locked to the given one?
$\mathcal{L}_G = 45\,\mathcal{L}_A^{w(f)} + \underbrace{100(\mathcal{L}_{\mathrm{IP}}+\mathcal{L}_{\mathrm{GD}}+\mathcal{L}_{\mathrm{IAF}})}_{\text{phase weight held at 100}} + 90\,\mathcal{L}_{\mathrm{com}} + 90\,\mathcal{L}_{\mathrm{cons}} + \mathcal{L}_{\mathrm{adv}} + \mathcal{L}_{\mathrm{fm}} + \underbrace{10\,\mathcal{L}_{\mathrm{SC}} + 3\,\mathcal{L}_{\mathrm{BE}} + 5\,\mathcal{L}_{\mathrm{LM}}}_{\text{amplitude level and shape 9}} + 1\,\mathcal{L}_{\mathrm{BPD}} + 0.2\,\mathcal{L}_{\mathrm{HPC}} + \underbrace{10\,\mathcal{L}_{\hat{\mathbf{d}}}}_{\text{blind encoder}}$
$\mathcal{L}_{\mathrm{SC}} = \||B| - |A|\|_F / \||B|\|_F$ (spectral convergence – relative, so a uniform gain error cannot hide) · $\mathcal{L}_{\mathrm{BE}}$ per-band energy in dB, ABOVE the cutoff only (the coarse-grained form of LSD) · $\mathcal{L}_{\mathrm{LM}}$ multi-resolution log-magnitude L_1 all three on the generator's own iSTFT output, at $(512, 128)$, $(1024, 256)$, $(2048, 512)$ · per-timestep weight $\lambda(t) = 1/c_{\mathrm{out}}(t)^2$ · $\mathcal{L}_{\mathrm{BPD}}$ is anti-wrapped baseband phase difference, with the deterministic $2\pi m H/N$ ramp removed · $w(f)$ ramps $1 \to 3$ above the cutoff
our contribution
trunk
quantum-inspired
standard DSP
1 blind degradation estimate
2 SBR prior: input channel, residual base and bridge endpoint
3 selective-SSM trunk, both streams
4 quantum-inspired amplitude modules: state-encoding features, and tensor-train factorisation of BOTH the input projection ($4.1\times$) and the output head ($3.8\times$)
5 Schrödinger bridge, $N{=}1$
6 phase path, held at the result that already matches AP-BWE's APD
7 amplitude-heavy critic set
8 baseband graft: the handed-over band is restored, not re-synthesised
9 amplitude level and shape losses
4 kHz → 16 kHz, VCTK, 6,250 steps on 2,000 files: LSD 1.20 · NISQA-MOS 3.39 · PESQ 2.20 · STOI 0.925 · SI-SDR 11.7 dB · APD 1.18 / 0.33 / 0.91